General Biology 1

First Semester

Supplementary Materials

Module 11

Cellular Abnormalities

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What I know

Directions: Read each situation carefully and choose the best answer.

1. What is the main cause of cancer at the cellular level?

a. Lack of oxygen in cells b. Uncontrolled cell division
c. Too much water in cells d. Slow metabolism

2. Which cellular problem is most associated with diabetes?

a. Cells divide too quickly
b. Cells cannot respond properly to insulin
c. Cells lose their nucleus
d. Cells produce too much oxygen

3. A scientist observes that a group of cells has mutations in genes that regulate the cell cycle. As a result, the cells continue to divide even when DNA is damaged. What is the most likely consequence of this abnormality?

a. The cells will stop dividing completely
b. The cells will repair themselves faster than normal
c. The cells may form tumors due to uncontrolled division
d. The cells will produce less energy

4. Why is studying diseases at the cellular level important?

a. It helps understand how diseases develop
b. It increases cell size
c. It prevents cell division
d. It changes DNA structure

1. A patient has cells that fail to undergo apoptosis even when they are severely damaged. What is the most likely outcome of this cellular abnormality?

a. Cells will divide more slowly than normal
b. Damaged cells may accumulate and form tumors
c. Cells will produce less DNA
d. The immune system will stop functioning

What I Need to Know

This supplementary module is designed to help you understand the **Cellular Abnormalities**. In this lesson, you will explore how cellular abnormalities lead to disease such as cancer, diabetes, and genetic disorders.

Lesson 11 Cellular Abnormalities

Cellular abnormalities are changes in a cell's structure, function, or genetic material that disrupt its normal behavior. These may result from mutations, environmental factors, or errors in cell division, leading to malfunctioning cells. Such abnormalities can contribute to diseases, including cancer, when cells grow uncontrollably or fail to function properly.

After going through this supplementary module, you are expected to:

1. identify and interpret information from secondary sources about cellular abnormalities related to diseases;
2. explain how specific cellular abnormalities, such as uncontrolled cell division, malfunctioning cells, and genetic mutations, lead to diseases such as cancer, diabetes, and genetic disorders; and
3. analyze the impact of these cellular abnormalities on body systems and evaluate their role in disease development and progression.

Activity 1: "Cells Gone Wrong: Investigating Disease Through Evidence"

Directions: Use information from secondary sources to explain how cellular abnormalities lead to diseases such as cancer, diabetes, and genetic disorders. Complete the table below.

Source 1: Cancer (Cell Division Problem)

Cancer occurs when cells divide uncontrollably due to mutations in genes that regulate the cell cycle. These abnormal cells form tumors and can spread to other parts of the body (metastasis). Normally, cell division is controlled, but cancer cells ignore these controls.

Source 2: Diabetes (Cell Signaling Problem)

Diabetes is a disease in which the body cannot properly regulate blood sugar levels. In Type 1 diabetes, the immune system destroys insulin-producing cells. In Type 2 diabetes, cells do not respond properly to insulin. This leads to high glucose levels in the blood.

Source 3: Genetic Disorder (Mutation Problem)

Genetic disorders are caused by mutations in DNA. For example, cystic fibrosis results from a defective protein that controls salt and water movement in cells. This causes thick mucus to build up in organs like the lungs.

Disease	Cellular Abnormality	Cause	Effect on the body
Cancer			
Diabetes			
Genetic Disorder			

Guide Questions:

1. How does a small change in a cell lead to a serious disease?

_____

_____

2. What do cancer, diabetes, and genetic disorders have in common at the cellular level?

_____

_____

3. Why is it important to understand diseases at the cellular level?

_____

_____

What's New

Activity 2: "Disease Detectives: Cracking Cellular Abnormalities"

Directions: Read each scenario carefully. Identify the cause of the disease by choosing: **Uncontrolled Cell Division**, **Malfunctioning Cells** and **Genetic Mutations**.

Scenario	Cause of Disease
A group of cells continues to divide rapidly even when the body does not need new cells, forming a tumor.	
A patient's cells cannot respond properly to insulin, causing high blood sugar levels.	
A patient's cells cannot respond properly to insulin, causing high blood sugar levels.	
Cells in the skin keep multiplying without stopping, leading to abnormal tissue growth.	
Cells in the skin keep multiplying without stopping, leading to abnormal tissue growth.	
A mutation in a gene causes the body to produce a defective protein that affects lung function.	
Cells ignore signals that regulate the cell cycle, continuing to divide uncontrollably.	
Muscle cells cannot function properly due to failure in energy production.	
A disorder is passed from parents to offspring due to an inherited DNA defect.	
A disorder is passed from parents to offspring due to an inherited DNA defect.	

Cellular abnormalities occur when normal cell processes are disrupted, and these changes can lead to various diseases. Three major types of abnormalities include uncontrolled cell division, malfunctioning cells, and genetic mutations, each affecting the body in different ways.

Uncontrolled cell division happens when cells divide continuously without proper regulation. Normally, the cell cycle is tightly controlled, but when this control fails—often due to damaged genes—cells keep multiplying. This leads to the formation of tumors and is the main cause of cancer, where abnormal cells can invade nearby tissues and spread throughout the body.

Malfunctioning cells occur when cells are present but do not perform their normal functions properly. For example, in diabetes, cells either do not produce enough insulin or do not respond to it correctly,

preventing glucose from entering the cells. As a result, blood sugar levels rise, affecting the body's ability to maintain balance and leading to long-term health problems.

Genetic mutations are changes in the DNA sequence that can alter how proteins are made or function. These mutations can be inherited or acquired over time. When important proteins are affected, normal cellular processes are disrupted. This can lead to genetic disorders such as cystic fibrosis, where a defective protein causes thick mucus to build up in the lungs and other organs.

What's More

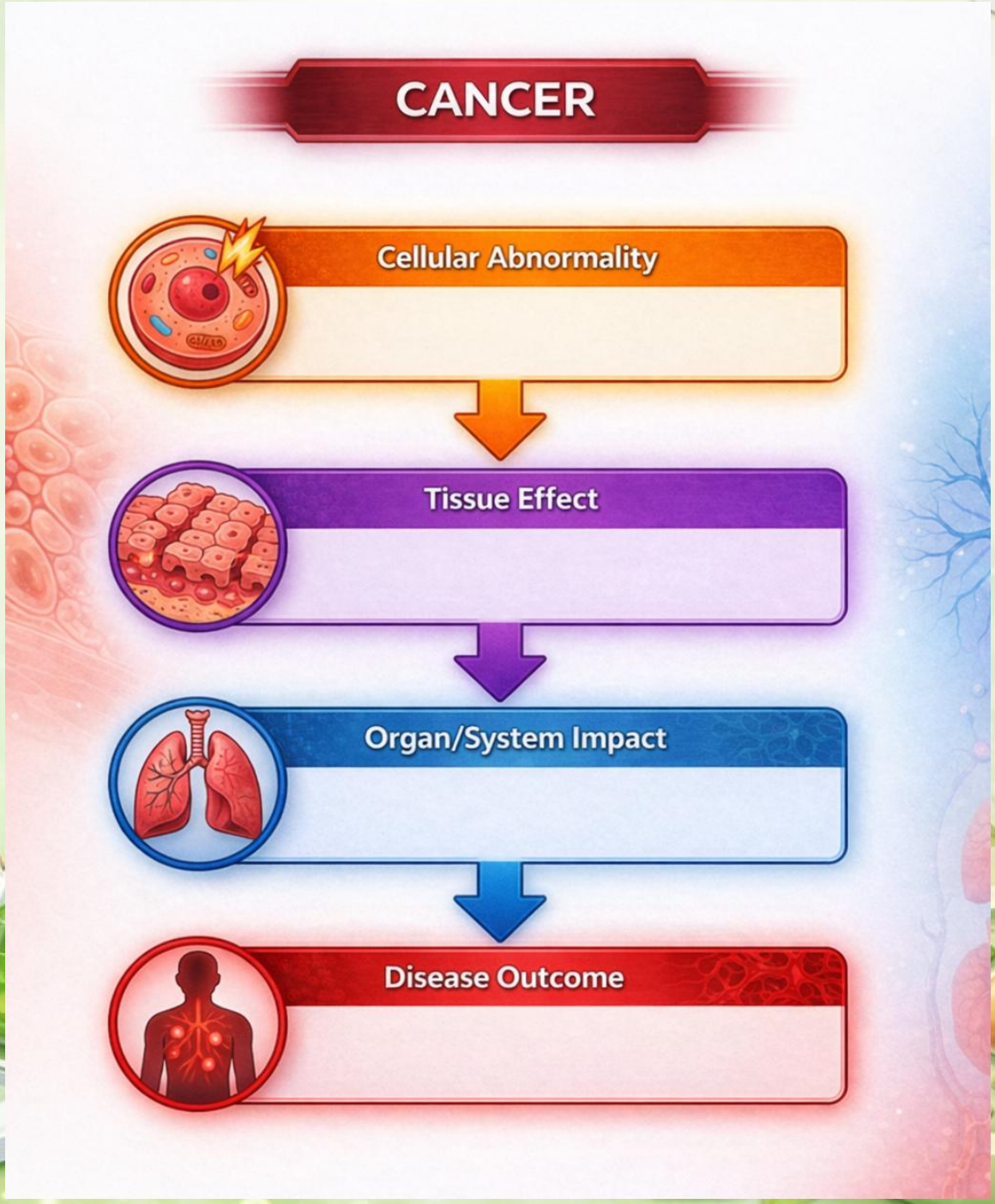
Activity 3: "Cellular Breakdown: From Cell Error to System Failure"

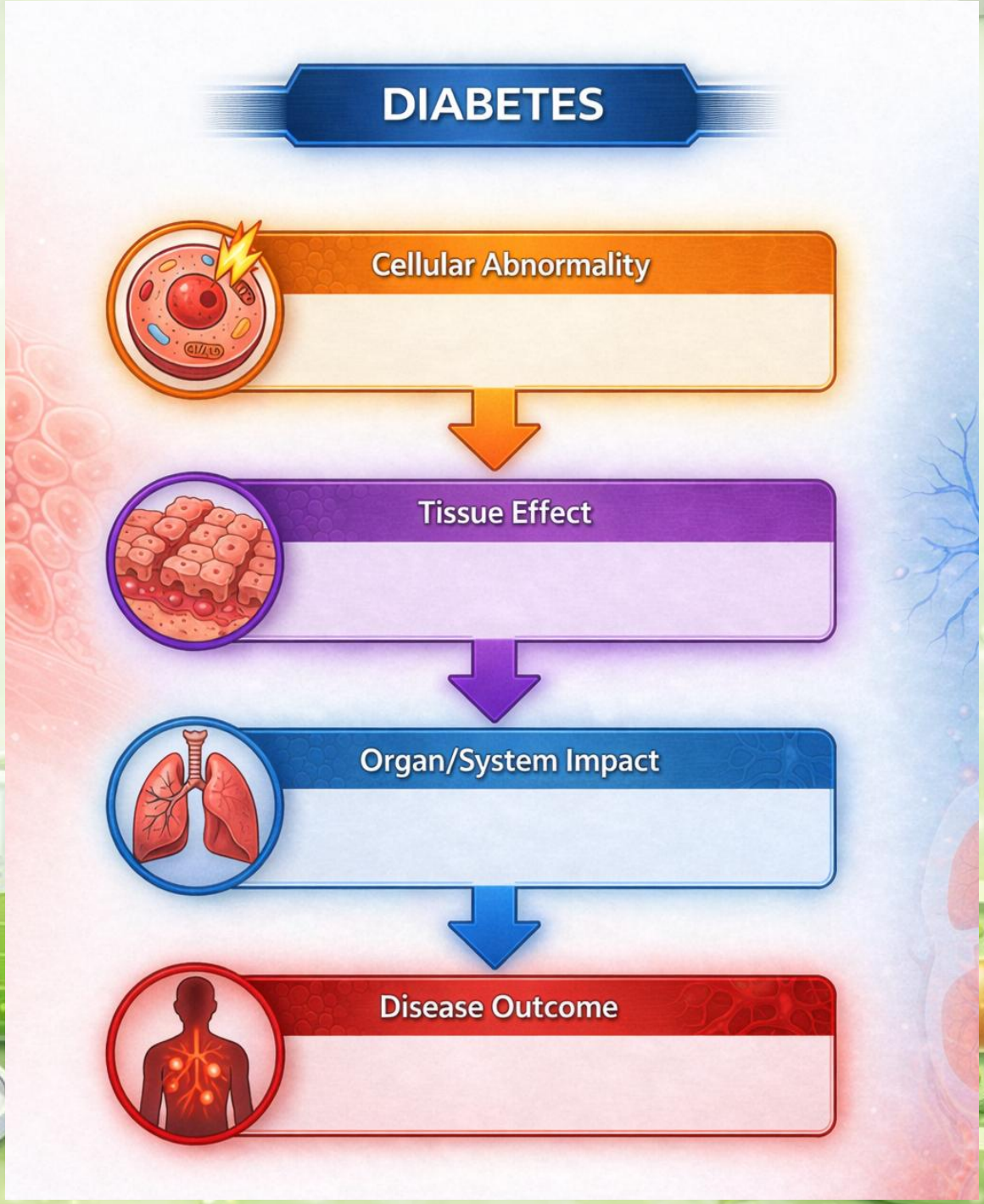
Directions: Analyze case studies, identify the type of cellular abnormality involved, and explain how it impacts specific body systems and contributes to disease by completing the flow chart in each disease.

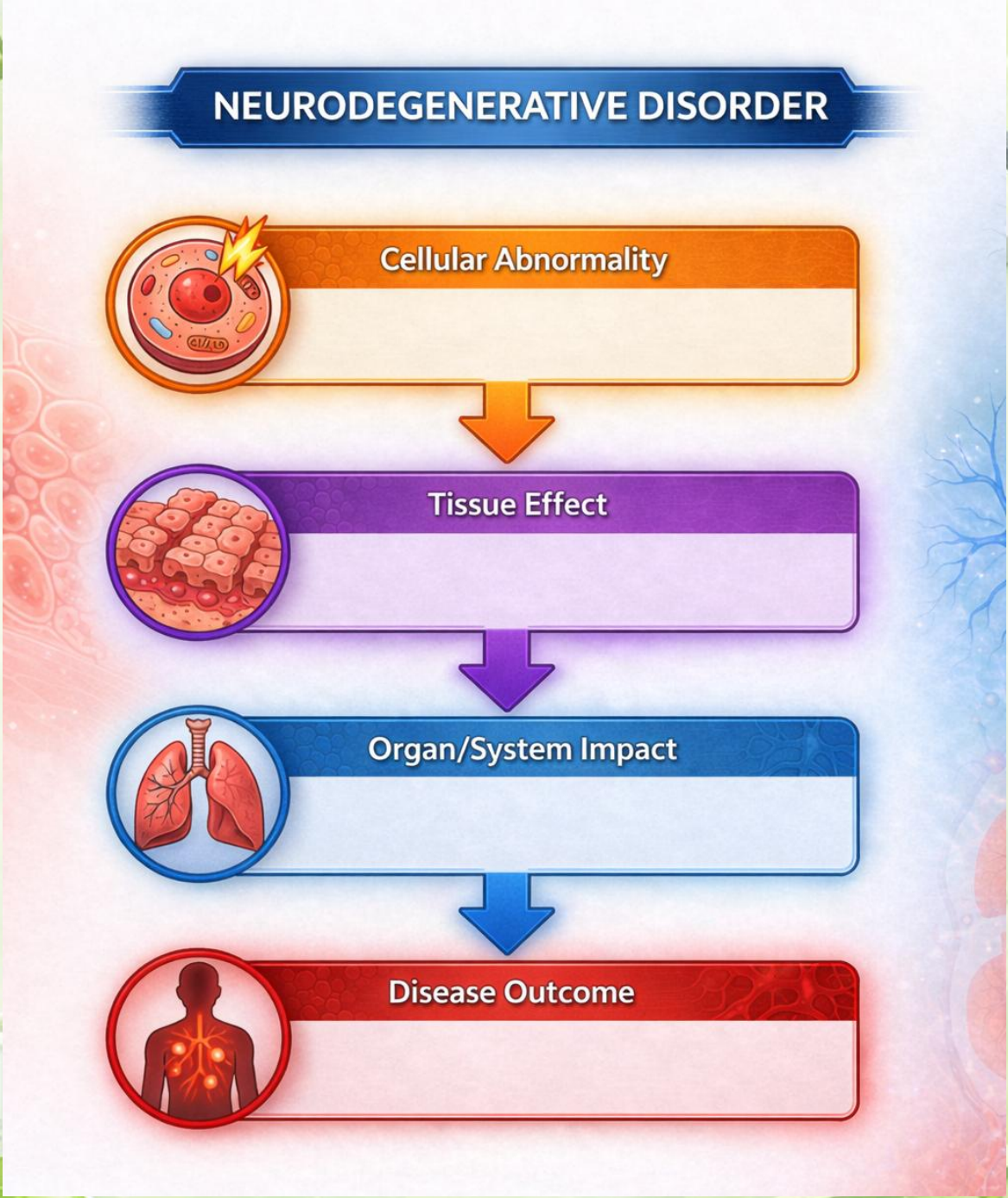
Activity Scenario

A hospital is investigating patients with different diseases such as cancer, diabetes, and neurodegenerative disorders. Doctors suspect that these conditions all begin with cellular abnormalities (e.g., uncontrolled cell division, malfunctioning organelles, genetic mutations).

Your group has been assigned as a medical investigation team to trace how these cellular problems affect entire body systems and lead to disease progression.


CANCER
Cellular Abnormality
Tissue Effect
Organ/System Impact
Disease Outcome


DIABETES
Cellular Abnormality
Tissue Effect
Organ/System Impact
Disease Outcome



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What I've Learned

Directions: Encircle the letter of the correct answer.

1. A tumor begins in lung tissue and later spreads to the brain and liver. What does this indicate about the cellular abnormality?
   a. Cells are functioning normally
   b. Cells are undergoing controlled division
   c. Cells are dividing uncontrollably and spreading (metastasis)
   d. Cells are only affecting one organ
2. A cell has a mutation that disables its ability to repair damaged DNA. What is the most likely long-term effect?
   a. The cell will stop dividing immediately
   b. The cell will produce more energy
   c. Mutations will accumulate, possibly leading to cancer
   d. The cell will become larger
3. How can malfunctioning insulin response in cells lead to damage in multiple body systems?
   a. It only affects the digestive system
   b. It prevents oxygen from entering cells
   c. It causes high blood sugar, which damages organs over time
   d. It increases cell division
4. A mutation affects a protein responsible for transporting substances in cells. What is the most likely result?
   a. Cells divide faster
   b. Substances cannot move properly, disrupting cell function
   c. Cells stop producing energy completely
   d. The membrane becomes thicker
5. Which statement best explains how a small cellular problem can affect an entire body system?
   a. Cells are independent of body systems
   b. One cell controls the entire body
   c. Cells form tissues and organs, so dysfunction spreads through the system
   d. Only large organs are affected by disease